

DNA Methylation Analysis Manual

Supplementary Materials — Group 4

MSc in Bioinformatics — University of Bologna

June 2025

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Preface

This document serves as an extensive technical guide for the DNA methylation analysis pipeline implemented by Group 4. It integrates theoretical foundations, detailed R code explanations, best practices, and troubleshooting tips to empower rigorous and reproducible Illumina HumanMethylation450K data analysis.

1 Glossary of Key Terms

Glossary (Part 1)

p-value The probability of obtaining a result as extreme as the observed one under the null hypothesis.

Detection p-value Metric indicating if the probe signal is distinguishable from background noise (values > 0.01 are often unreliable).

Beta-value Ratio of methylated to total signal, representing methylation level (range 0–1).

M-value $\log_2$ ratio of methylated to unmethylated signal; preferred for statistical testing due to improved variance properties.

CpG site DNA region where a cytosine nucleotide is followed by a guanine; key location for DNA methylation.

Heatmap A color-coded matrix visualization representing probe methylation across samples, often used for clustering analysis.

Workspace The R environment that stores all loaded objects during a session.

Object Any data structure stored in R, such as vectors, data frames, or lists.

Function A block of R code that performs a specific task.

Variable A name assigned to an object in R, enabling later reference.

Script A text file (.R) containing a sequence of R commands.

Console The interactive R prompt in RStudio or R.

Data frame A two-dimensional table-like structure with rows and columns.

Vector A one-dimensional sequence of data elements of the same type.

List A flexible container that can store multiple types of R objects.

False Positives Results that appear significant by chance but do not reflect true biological effects.

Benjamini-Hochberg (BH) A method for controlling the false discovery rate (FDR) when performing multiple statistical tests.

Bonferroni A conservative correction method for multiple testing that controls the family-wise error rate by adjusting p-values.

PC number Principal Component (PC) number; indicates the order of variance explained by each component in PCA, where PC1 captures the most variance.

Variance A statistical measure of data spread; in PCA, it represents how much variation each principal component explains.

Linkage In hierarchical clustering, linkage defines how distances between clusters are calculated; methods include complete, average, and single linkage.

Glossary (Part 2)

Fold Change The difference in mean Beta-values between two groups (e.g. disease vs. control), indicating the magnitude of methylation change.

QC (Quality Control) A set of procedures to ensure the reliability and accuracy of experimental data before downstream analysis.

Loop A control structure that repeats a block of code multiple times.

Operator A symbol (e.g., +, -, >, <, ==) used to perform operations.

False Discovery Rate (FDR) Correction method that controls the expected proportion of false positives among significant results.

Dye Bias Technical artifact introduced by different dye efficiencies; corrected via normalization.

Manifest file Metadata file that maps probe IDs to genomic coordinates and annotations.

RGChannelSet minfi object storing raw red and green channel intensities.

MethylSet minfi object after background correction, ready for downstream analysis.

Normalization Process that adjusts data to remove technical variation while preserving biological differences.

Batch effect Unwanted variation introduced by experimental conditions (e.g., processing date, operator).

Pipeline An ordered sequence of analysis steps.

Preprocessing Steps like background correction and normalization applied to raw data to prepare it for analysis.

Differential methylation Statistical comparison of methylation levels between groups (e.g., disease vs. control).

t-test A statistical test comparing the means of two groups to identify significant differences.

Volcano plot A scatter plot showing significance ($-\log_{10}$ p-value) versus effect size (e.g., delta beta).

Manhattan plot Genome-wide plot showing $-\log_{10}$ p-values by chromosome position.

PCA (Principal Component Analysis) A technique to explore sample variance and detect batch effects.

Pipe operator (%>%) An operator from magrittr/dplyr to chain commands for cleaner syntax (not used in this workflow).

2 Theoretical Background

DNA Methylation

DNA methylation is an epigenetic modification that occurs when a methyl group is added to the 5' position of cytosine residues within CpG dinucleotides. This modification is crucial for regulating gene expression and maintaining genome stability. Aberrant methylation is associated with diseases such as cancer and neurological disorders.

Illumina 450K BeadChip

The Illumina HumanMethylation450K BeadChip enables high-throughput analysis of over 485,000 CpG sites across the genome. It uses bisulfite conversion to differentiate between methylated and unmethylated cytosines:

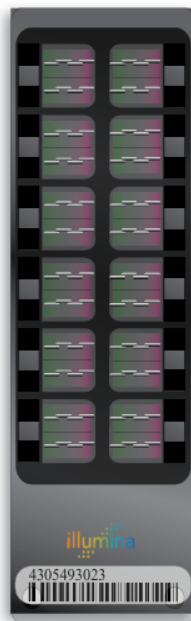
- **Infinium I:** Utilizes two separate probes per CpG site—one specific for the methylated sequence and one for the unmethylated sequence. Each probe uses single-color detection in separate bead types.
- **Infinium II:** Utilizes a single probe per CpG site that hybridizes regardless of methylation status. The methylation state is distinguished using two-color detection in the same bead type.

Bisulfite Conversion

Bisulfite treatment converts unmethylated cytosines into uracils (read as thymines after PCR), while methylated cytosines remain unchanged. This differential conversion allows the array probes to detect methylation states.

Characteristic	Infinium I	Infinium II
Number of Probes per CpG Site	Two separate probes (one for methylated, one for unmethylated)	One single probe detects both methylated and unmethylated sequences
Probe Design	Two different bead types for methylated and unmethylated targets	One bead type per CpG site hybridizes regardless of methylation status
Single Base Extension (SBE)	Adds labeled nucleotide at the end of each probe; separate color channels (G = methylated, A = unmethylated)	Adds labeled nucleotide at the end of the single probe; both signals detected in the same color channel (G = methylated, A = unmethylated)
Detection Channel	Single color channel detects both methylated and unmethylated	Two separate color channels for methylated and unmethylated probes
CpG Coverage	Slightly lower overall coverage due to array space used by dual probes	Slightly higher overall coverage due to single-probe design
Signal Intensity and Dynamic Range	Higher dynamic range; less compression of Beta values	Lower dynamic range; more compression of Beta values
Dye Bias Susceptibility	Less dye bias, easier normalization	Higher dye bias, requires adjustment
Data Interpretation	Raw signals easier to interpret directly	Requires correction for dye bias and sequence context

Table 1: Comparison of Infinium I and Infinium II probe chemistries used in Illumina DNA methylation BeadChip arrays.



The Infinium HumanMethylation450 BeadChip features more than 450,000 methylation sites, within and outside of CpG islands.

Figure 1: Overview of the Illumina Infinium HumanMethylation450 BeadChip platform. This array enables high-throughput analysis of over 485,000 methylation sites per sample at single-nucleotide resolution. It covers 99% of RefSeq genes, targeting an average of 17 CpG sites per gene region (including promoter, 5'UTR, first exon, gene body, and 3'UTR). The platform also covers 96% of CpG islands, as well as island shores and flanking regions. Additional features include coverage of non-CpG methylated sites (identified in human stem cells), differentially methylated sites between tumor and normal samples (multiple cancer types), CpG islands outside of coding regions, and miRNA promoter regions. These features make the platform ideal for epigenome-wide association studies (EWAS).

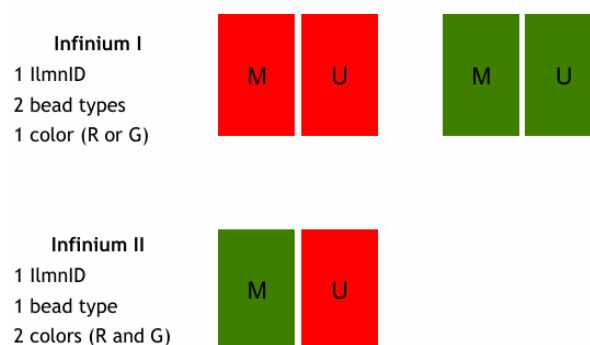


Figure 2: Comparison of Infinium I and Infinium II probe designs.

3 R — A Deep Dive

What is R?

R is a free, open-source programming language and environment specifically designed for statistical computing and data visualization. It is widely used in bioinformatics because of its:

- Flexibility in handling large and complex datasets typical of omics studies.
- Extensive ecosystem of packages dedicated to genomics and epigenetics (e.g. `minfi`, `limma`, `edgeR`).
- Integration with Bioconductor, providing standardized workflows for methylation analysis.
- Seamless interface with RStudio, an integrated development environment (IDE) that simplifies script writing, data visualization, and debugging.

R is particularly valued for its ability to chain together data preprocessing, statistical testing, visualization, and result export, making it a complete solution for DNA methylation analysis.

The RStudio Interface

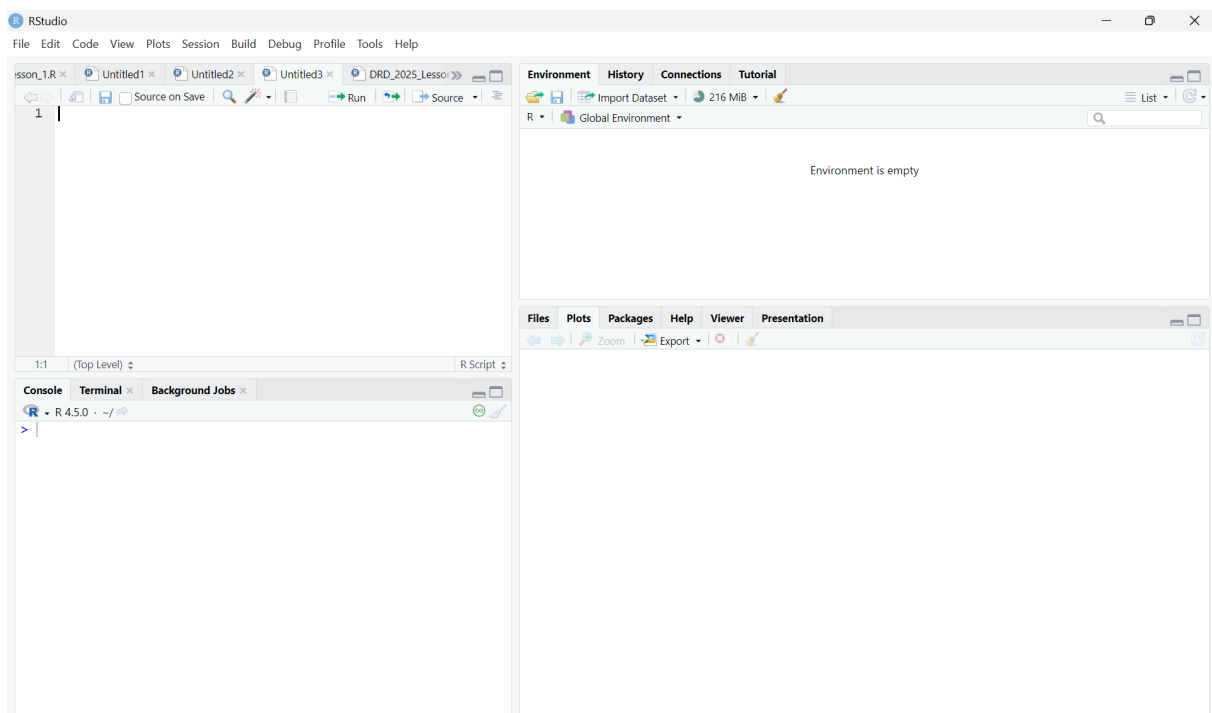


Figure 3: RStudio Interface — Console, Script Editor, Workspace, Plots.

Tips and Useful Commands in R

To efficiently work with R, it is helpful to know some basic tips and commands:

- Use the assignment operator `<-` to create objects: `mydata <- read.csv("data.csv")`.
- Clean the workspace to avoid object conflicts: `rm(list = ls())`.
- Use the `#` symbol to comment and annotate code.
- Check installed packages with `installed.packages()` and load them with `library(pkgname)`.
- Explore data with commands like `str()`, `summary()`, `head()`, and `tail()`.
- Use pipes (`%>%`) from the `magrittr` or `dplyr` package to chain commands and improve code readability.
- For quick plotting, try `plot()`, while for more advanced graphics use `ggplot2`.

Packages and Libraries

Packages extend R's functionality. Use:

Installing and Loading Packages

```
install.packages("ggplot2")
library(ggplot2)
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
BiocManager::install("minfi")
```

Data Structures in R

In R, data structures are fundamental for organizing and analyzing data. Here are the most common structures you will encounter in methylation analysis:

- **Vector:** A one-dimensional sequence of elements of the same type (numeric, character, logical).
- **Factor:** A special type of vector for categorical data with predefined levels.
- **Matrix:** A two-dimensional array of elements of the same type.
- **Data Frame:** A table-like structure with rows and columns, where each column can have a different type.
- **List:** A flexible container that can hold objects of different types and structures (e.g. data frames, vectors, even other lists).

Example:

```
# Vector
my_vector <- c(1, 2, 3, 4)

# Factor
my_factor <- factor(c("Control", "Treatment", "Control"))
```

```
# Data Frame
my_df <- data.frame(SampleID = c("A1", "A2"), Value = c(0.5, 0.7)
)

# List
my_list <- list(Numeric = my_vector, Category = my_factor, Table
= my_df)
```

Understanding these data structures is essential for efficient data manipulation and analysis in R.

4 About minfi

The **minfi** package is a comprehensive and essential R/Bioconductor toolkit for analyzing Illumina HumanMethylation arrays (450K and EPIC). It enables efficient processing and robust analysis of DNA methylation data by providing:

- Import of raw IDAT files using `read.metharray.exp()`, supporting flexible sample sheet integration.
- Quality control metrics, such as probe detection p-values via `detectionP()` and negative control plots, to ensure data reliability.
- Preprocessing methods, including background correction and functional normalization (`preprocessFunnorm()`), to correct technical biases and batch effects.
- Extraction of Beta-values and M-values using `getBeta()` and `getM()`, facilitating downstream statistical testing.

Overall, **minfi** supports a complete methylation analysis workflow, from raw data import to preprocessing and quality assessment, making it an essential package for reproducible and high-quality methylation studies.

Note. Additional R packages were also used to support data visualization and specialized analyses, including:

- **gplots** for heatmaps and hierarchical clustering.
- **qqman** for generating Manhattan plots.
- **factoextra** for PCA visualization.

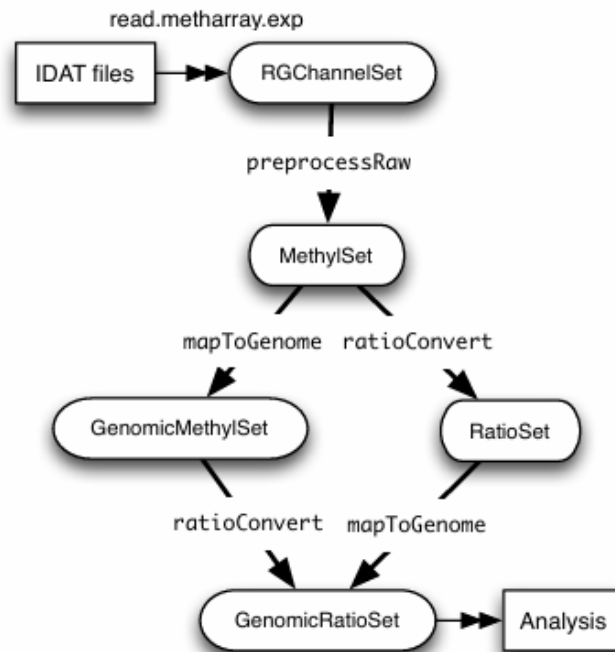


Figure 2: Flow chart of the `minfi`'s functions

Figure 4: Flow chart of the `minfi` pipeline: data objects evolve from raw IDAT files (`RGChannelSet`) to genome-annotated methylation matrices (`GenomicRatioSet`). Each transformation adds structure and annotation, enabling quality control, normalization, and biological interpretation.

Each object plays a defined role within the `minfi` pipeline:

- **`RGChannelSet`**: Raw intensity data (red and green channels) from IDAT files, unfiltered and unprocessed.
- **`MethylSet`**: Background-corrected methylated and unmethylated signals, still unmapped to the genome.
- **`RatioSet`**: Includes Beta-values (methylated / total) and/or M-values ($\log_2$ ratio), useful for differential methylation analysis.
- **`GenomicMethylSet`**: Like `MethylSet`, but with genome mapping information.
- **`GenomicRatioSet`**: Final analysis-ready object: Beta or M-values mapped to genome coordinates, ideal for visualization and statistical testing.

Function name	Package	Input	Output	Ref
preprocessNoob	minfi	RGset	Mset	Triche 2013
preprocessSWAN	minfi	RGset	Mset	Maksimovic 2012
preprocessQuantile	minfi	RGset or Mset	Genomic Ratio Set	Toulemait 2012
preprocessFunnorm	minfi	RGset	Mset	Fortin 2014

Figure 5: Comparison of preprocessing functions available in the `minfi` package. Each method takes `RGset` (raw data) as input and produces methylation-level objects (`Mset` or `GenomicRatioSet`).

Theoretical note: Several preprocessing strategies are available within `minfi`, each tailored to different correction goals:

- `preprocessNoob` corrects background and dye bias (single-sample method).
- `preprocessSWAN` adjusts for probe type bias between Infinium I and II probes.
- `preprocessQuantile` equalizes the distribution across samples, often used in general omics.
- `preprocessFunnorm` uses internal control probes to remove unwanted variation while preserving biological signal — this was the method used in our analysis.

5 Project Data

For this project, we analyzed DNA methylation data from the Illumina HumanMethylation450K array. Below is a summary of the key files and data objects used in the analysis:

File/Data Object	Description
SampleSheet.csv	Sample metadata including ID, group (CTR-L/DIS), sex, and batch.
.idat files	Raw fluorescence intensity data (red and green channels).
Illumina450Manifest_clean.RData	Cleaned manifest with probe IDs, genomic coordinates, and design type.
RGChannelSet	Raw data object created with <code>read.metharray.exp()</code> .
MSet.raw	Preprocessed object after background correction (<code>preprocessRaw()</code>).
MSet.norm	Normalized data using functional normalization (<code>preprocessFunnorm()</code>).
Beta/M values	Matrices of methylation quantification (<code>getBeta()</code> , <code>getM()</code>).
DetectionP	Detection p-values per probe/sample (<code>detectionP()</code>).
final_ttest	Data frame with raw p-values from group comparison.
final_ttest_corr	Data frame with corrected p-values (BH and Bonferroni).
diff_meth_df	Summary table of differentially methylated probes.
pca_results	PCA results object for sample clustering.
Plots	Figures including density plots, boxplots, PCA, volcano, Manhattan plots, and heatmaps.

Table 2: Key files and data objects used in the Group 4 DNA methylation analysis.

Additionally, the following **assigned parameters** were used for the analysis:

Parameter	Value
Group ID	4
Probe Address	44666390
Detection p-value cut-off	0.01
Normalization method	<code>preprocessFunnorm()</code>
Clustering methods	Average, Complete, Single linkage

Table 3: Assigned parameters used in the analysis.

6 Pipeline Overview

An expanded step-by-step pipeline:

1. **Data Import:** Read raw .idat files using `read.metharray.exp()` to create an `RGChannelSet`.

2. **Annotation:** Use the cleaned manifest file (`Illumina450Manifest_clean.RData`) to map probe IDs and design types.
3. **Quality Control:** Calculate detection p-values (`detectionP()`) and generate quality control plots (`plotQC()`, `controlStripPlot()`).
4. **Preprocessing:** Apply background correction with `preprocessRaw()` and functional normalization with `preprocessFunnorm()`.
5. **Beta/M-value Calculation:** Calculate Beta-values (`getBeta()`) and M-values (`getM()`).
6. **Statistical Testing:** Perform t-tests (`t.test()`) comparing groups.
7. **Multiple Testing Correction:** Apply correction for multiple testing using `p.adjust()` (methods: BH, Bonferroni).
8. **Visualization:** Generate density plots, boxplots, PCA plots (`prcomp()`, `fviz_eig()`), volcano plots, Manhattan plots (`qqman`), and heatmaps (`heatmap.2()`).

7 The Analytical Pipeline — From Biological Question to Visualization

A successful DNA methylation study requires a well-designed workflow that systematically addresses every analytical step. Below, we outline a typical pipeline, emphasizing its logical flow from hypothesis formulation to data visualization:

1. Biological Question

- Define the scientific goal, e.g., “Does methylation differ between disease and control groups?”
- Establish hypotheses and experimental design (number of samples, replicates, conditions).

2. Sample Collection and Preparation

- Extract DNA from relevant tissues or cell types.
- Perform bisulfite conversion to differentiate methylated from unmethylated cytosines.

3. Data Acquisition

- Use the Illumina HumanMethylation450K array (or EPIC) to profile methylation.
- Generate raw data files (`.idat`) containing red and green channel intensities.

4. Data Import into R

- Use `read.metharray.exp()` to read `.idat` files into an `RGChannelSet` object.
- Inspect sample metadata (e.g., group, sex, batch) from `SampleSheet.csv`.

5. Annotation

- Link probe IDs to genomic features using the cleaned manifest (`Illumina450Manifest_clean.RData`).
- Exclude problematic probes (e.g., SNP-overlapping, cross-hybridizing).

6. Quality Control

- Calculate detection p-values with `detectionP()` to filter unreliable probes.
- Assess sample quality using `plotQC()` and `controlStripPlot()` for negative controls.

7. Preprocessing

- Apply background correction using `preprocessRaw()`.
- Normalize data using functional normalization with `preprocessFunnorm()` to mitigate technical biases and batch effects.

8. Methylation Quantification

- Calculate Beta-values with `getBeta()` and M-values with `getM()`.

9. Statistical Testing

- Perform t-tests for group comparisons (`t.test()`).

10. Multiple Testing Correction

- Correct p-values for multiple testing using `p.adjust()` (methods: BH and Bonferroni).

11. Visualization

- Generate density plots, boxplots, PCA plots (`prcomp()`, `fviz_eig()`), volcano plots, Manhattan plots (`qqman`), and heatmaps (`heatmap.2()`).

12. Biological Interpretation

- Map significant CpGs to genes and pathways.
- Integrate methylation findings with external datasets (e.g., gene expression, clinical data).

This pipeline illustrates how each step builds upon the previous one, transforming raw experimental data into biologically meaningful insights.

8 Interpretation of Outputs and Parameter Choices

A high-level technical discussion that complements the pipeline described earlier, focusing on the rationale behind key analytical choices and their impact on data quality, statistical reliability, and interpretation of results. By integrating theoretical principles with observations from the outputs, this discussion bridges the analysis steps with the biological and technical context of the study.

8.1 Parameter Choices and Data Quality

Functional Normalization (`preprocessFunnorm()`) was selected to correct dye bias, background noise, and partial batch effects, leveraging internal control probes to harmonize data across samples. This choice preserves true biological variability while minimizing technical noise.

Detection p-value Cutoff (0.01) provided a stringent filter to exclude unreliable probes whose signals may not exceed background noise. Although conservative, this threshold ensures that only high-confidence measurements are included, supporting reliable downstream analyses.

8.2 Quality Control and Normalization

Negative control and QC plots confirmed consistently low background fluorescence across samples, indicating that no sample exhibited technical anomalies. Density plots comparing raw and normalized data demonstrated that normalization effectively aligned intensity distributions, correcting systematic biases and enabling accurate group comparisons.

8.3 Methylation Distributions

Beta-value and M-value density plots displayed the expected bimodal distributions, indicating clear separation between methylated and unmethylated states and confirming the biological relevance of the data.

8.4 Principal Component Analysis (PCA)

PCA plots revealed clear separation between control and disease groups, with minimal batch and sex effects, demonstrating that potential confounding variables were adequately controlled.

8.5 Differential Methylation and Multiple Testing

The volcano plot highlighted CpGs with substantial methylation differences and significant adjusted p-values, supporting the presence of biologically relevant differential methylation. The p-value distribution plot confirmed a uniform distribution under the null hypothesis, with enrichment at low p-values consistent with true biological signals.

P-Value Boxplots (Raw, BH, Bonferroni)

Boxplots comparing raw, BH-adjusted, and Bonferroni-adjusted p-values illustrated the impact of multiple testing correction: while many CpGs appeared significant using raw p-values, only a subset remained significant after correction, emphasizing the importance of controlling false discovery rates.

Note: In this analysis, a nominal p-value threshold of 0.05 was adopted for the initial t-tests, even though the detection p-value cutoff was set at 0.01 during the quality control phase. This nominal threshold (0.05) is a conventional standard in hypothesis testing that balances sensitivity and specificity prior to applying multiple testing corrections (BH and Bonferroni). The detection p-value cutoff (0.01), on the other hand, is used earlier in the

pipeline to filter out unreliable probes and is not directly linked to the statistical threshold applied in the differential testing step.

8.6 Genome-Wide Overview

The Manhattan plot visualized genome-wide significance, showing clusters of differentially methylated CpGs at specific chromosomal locations, supporting downstream biological interpretation.

8.7 Clustering and Heatmaps

Heatmaps using complete, average, and single linkage methods consistently grouped samples by biological condition (CTRL vs. DIS). The color gradient from green to red effectively represented the continuum from hypomethylation to hypermethylation.

8.8 Limitations and Potential Improvements

While the chosen parameters provided a strong analytical foundation, several refinements could be explored. A detection p-value cutoff of 0.01 is conservative and may exclude borderline CpGs with potential biological relevance; testing a slightly higher threshold (e.g., 0.05) could help balance stringency and inclusiveness. Alternative normalization methods, such as quantile normalization or SWAN, might perform differently depending on dataset characteristics. Although batch effects appeared minimal in PCA, additional correction using methods like ComBat could be tested to further reduce potential technical variance. Exploring different probe selections or group assignments might also assess the robustness of the findings under alternative analytical settings.

9 Function Reference

Data Import

```
read.metharray.exp() | Imports raw .idat files into an RGChannelSet object.
```

Annotation

```
Illumina450Manifest_clean | Cleaned manifest file for probe annotation.
```

Quality Control

```
detectionP() | Computes detection p-values per probe.  
plotQC() | Visualizes sample quality metrics.  
controlStripPlot() | Visualizes negative control probes.
```

Preprocessing

```
preprocessRaw() | Applies background correction.  
preprocessFunnorm() | Performs functional normalization.
```

Beta/M-value Calculation

```
getBeta() | Calculates Beta-values.  
getM() | Calculates M-values.
```

Statistical Testing

```
t.test() | Tests for group differences.
```

Multiple Testing Correction

```
p.adjust() | Adjusts p-values for multiple testing (BH, Bonferroni).
```

Visualization

```
heatmap.2() | Creates heatmaps with clustering.  
fviz_eig() | Plots PCA eigenvalues.  
manhattan() | Generates Manhattan plots.
```

10 Tips and Troubleshooting

- **Object Not Found:** Check spelling carefully (case-sensitive) and ensure that the object was created in the current R session. Use `ls()` to list all objects in the workspace.
- **Package Not Loaded:** Always load packages with `library()` before using their functions. For Bioconductor packages, remember to install via `BiocManager::install()` first.
- **Dimension Mismatch:** Check that data frames or matrices used in functions have the same number of rows/columns or match the expected structure. Use `dim()` and `head()` to inspect.
- **Warning Messages:** R issues warnings (e.g. `Warning: NAs introduced by coercion`) that do not stop code but may indicate potential problems. Review warnings carefully with `warnings()` and adjust code accordingly.
- **Missing Values (NAs):** Check for missing values with `is.na()` and decide on appropriate handling (e.g. remove or impute).

- **Normalization Issues:** If downstream analyses seem off (e.g. unexpected sample clustering or dye bias), verify that background correction and normalization have been correctly applied. Re-check steps like `preprocessRaw()` and `preprocessFunnorm()`.
- **Dye Bias:** A technical artifact especially in Infinium II probes. Use appropriate normalization methods to correct (e.g. functional normalization).
- **Version Conflicts:** Make sure R and packages are updated to avoid deprecated functions. Check package version with `packageVersion("packagename")`.
- **Visualization Errors:** If plots look wrong or incomplete, check that the data is correctly structured and complete. Use `ggplot2` syntax carefully.
- **Unexpected Results:** Always verify that the correct samples and group labels were assigned during preprocessing. Use `table()` to cross-check factors like group and batch.

11 Conclusion

This manual is designed to guide users through the process of DNA methylation data analysis, from data import to visualization and interpretation. Our project demonstrates the critical role of bioinformatics in modern molecular biology and highlights the power of reproducible research. By using robust tools such as R and the minfi package, we can process large datasets efficiently and gain valuable insights into epigenetic regulation and disease mechanisms.

The expected results include the identification of differentially methylated regions (DMRs) that may serve as biomarkers for disease, as well as the discovery of new epigenetic patterns associated with clinical outcomes. This work contributes to the broader goal of personalized medicine, where DNA methylation signatures can inform both diagnostics and therapeutic strategies.

We hope this document empowers other researchers to replicate our pipeline and adapt it to their own datasets, fostering collaboration and accelerating discoveries in the field of epigenetics.